Wesley DeMontigny

Computational Scientist | Bayesian & Probabilistic Modeling

wes.demontigny@gmail.com, GitHub, Portfolio

Education:

Ph.D. Candidate in Biological Sciences, *May 2027*, *University of Maryland* GPA: 4.0

- **Focus**: Computational Statistics, Bayesian Inference, and Computational Biology
- **Dissertation Topic**: Imputation Models for Partially Observed Genomic Datasets

B.S. in Biology, *2022*, *Georgia College & State University* GPA: 3.95

- **Awards**: Goldwater Scholar (2022) | Commencement Speaker for College of Arts & Sciences

Graduate Research Assistant, University of Maryland | 2023–Present

- Developed probabilistic models and computational methods for inference from incomplete, heterogeneous biological datasets.
- Built end-to-end computational pipelines for assembly, annotation, and comparative analysis of eukaryotic genomes from cultured and environmental samples.

Selected Projects:

Probabilistic Imputation Model for Incomplete Genomic Data | GitHub, Preprint (under review)

- Designed and implemented a probabilistic graphical model in Python (JAX/NumPyro) to infer missing gene content in incomplete genomic datasets. Released the model as a reusable Python package for downstream genomic analysis.
- Demonstrated robustness to misspecification, with minimal performance degradation under incorrect assumed relationships.
- Achieved 2× higher precision than competing methods at comparable recall, while providing uncertainty quantification for downstream analysis.
- Extended the framework to model source-specific detection errors data derived from different procedures and to infer protein functional associations (*Ongoing*).

Bayesian Modeling of Protein Evolution | GitHub, Preprint (under review)

- Engineered a high-performance C++17 reversible-jump MCMC sampler for a nonparametric Bayesian model that infers site-specific selective pressures and maps evolutionary constraint onto protein structure.
- Ran large-scale simulation studies on HPC clusters using Slurm and validated sampler correctness with posterior coverage tests.
- Applied the model to hemoglobin evolution, mapping site-specific selection onto 3D protein structures and recovering clinically relevant residues that distinguish paralogs.

International Forecasting Competition — 1st Place / Most Accurate Prediction (2026) | Write-Up

International Cherry Blossom Prediction Competition, George Mason Statistics

- Produced the lowest forecast error across five international sites using mixed-effects models and learned LSTM representations of climate time series.
- Pre-trained a stacked PyTorch LSTM on NASA POWER temperature and NOAA ENSO time series; transferred learned representations to an MLP for forecasting with limited labeled data.

Scientific Software Development | Various Projects

- Diagnosed and fixed critical defects in a legacy C++ Bayesian MCMC codebase and implemented automatic proposal tuning, improving sampling efficiency by >2× and enabling completion of a previously stalled research project.
- Developed custom proposal distributions and adaptive tuning strategies to accelerate MCMC convergence in high-dimensional, strongly correlated posterior spaces.

The Bayesian Battleship Benchmark | GitHub, Write-Up

- Designed a Sequential Monte Carlo sampler to approximate the combinatorial posterior over constrained Battleship configurations.
- Benchmarked Bayesian decision policies (MAP, information gain, Thompson sampling) against frontier and open-source LLMs, quantifying performance via Bradley–Terry skill models, posterior deviation, and information-gain metrics.

Relevant Publications (* Denotes Co-First Authorship):

Mattick J.S.A., ***DeMontigny W.C.**, and Delwiche C.F. Inferring Gene Presence in Incomplete Data via Phylogenetic Occupancy Modeling. *bioRxiv*.

DeMontigny W.C. and Delwiche C.F. OmegaSwitch: Bayesian Markov Modulated Codon Models for Estimating *dN/dS*. *bioRxiv*.

DeMontigny W.C. and Bachvaroff, T. The nuclear and mitochondrial genomes of *Amoebophrya* sp. ex *Karlodinium veneficum*. *G3: Genes, Genomes, and Genomics, 15 (2025).*

Seah B.K.B., Shaikhutdinov N., **DeMontigny W.C.** et al. Spliceosome loss in the red tide ciliate *Mesodinium rubrum* presents a symbiotic cul-de-sac. *bioRxiv*.

Skills:

- **Languages & Tools:** C++, Python, R, SQL, Linux CLI, Bash, Git, CMake, Docker, Slurm
- **Scientific Computing & ML Libraries:** PyTorch, PyTorch Geometric, JAX, NumPyro, scikit-learn, NumPy, SciPy, pandas, Eigen, OpenMP, Boost
- **Modeling**: Bayesian inference, probabilistic graphical models, MCMC/SMC, latent-variable models, time-series modeling, nonparametric methods, deep learning.